Johns Hopkins Engineering

Software Radio for Wireless Communications

Module 12 Live Lab

JOHNS HOPKINS
WHITING SCHOOL
of ENGINEERING

Goals

- This Live Lab will:
  - Develop skills with
    - Convolutional Neural Nets (CNNs) for Modulation Classification
    - Hidden Markov Models (HMMs) for predicting activity

Outline

1. Questions from Last Week’s Homework
2. Brief Discussion on Secondary Users and Whitespace
3. Background on HMMs
4. Experiments
5. Background on CNNs
6. Experiments

Last Week’s Homework

Last Week’s Homework

- Q & A

Hidden Markov Models and Channel Activity Prediction

Secondary Spectrum Access (1)

- Secondary spectrum access (SSA)
  - Primary user (PU) retains priority in accessing resources
  - Secondary user (SU) accesses resources when available

- SSA maximizes spectrum utilization while maintaining PU primacy
  - In theory … <u>this is not a legal construct</u>

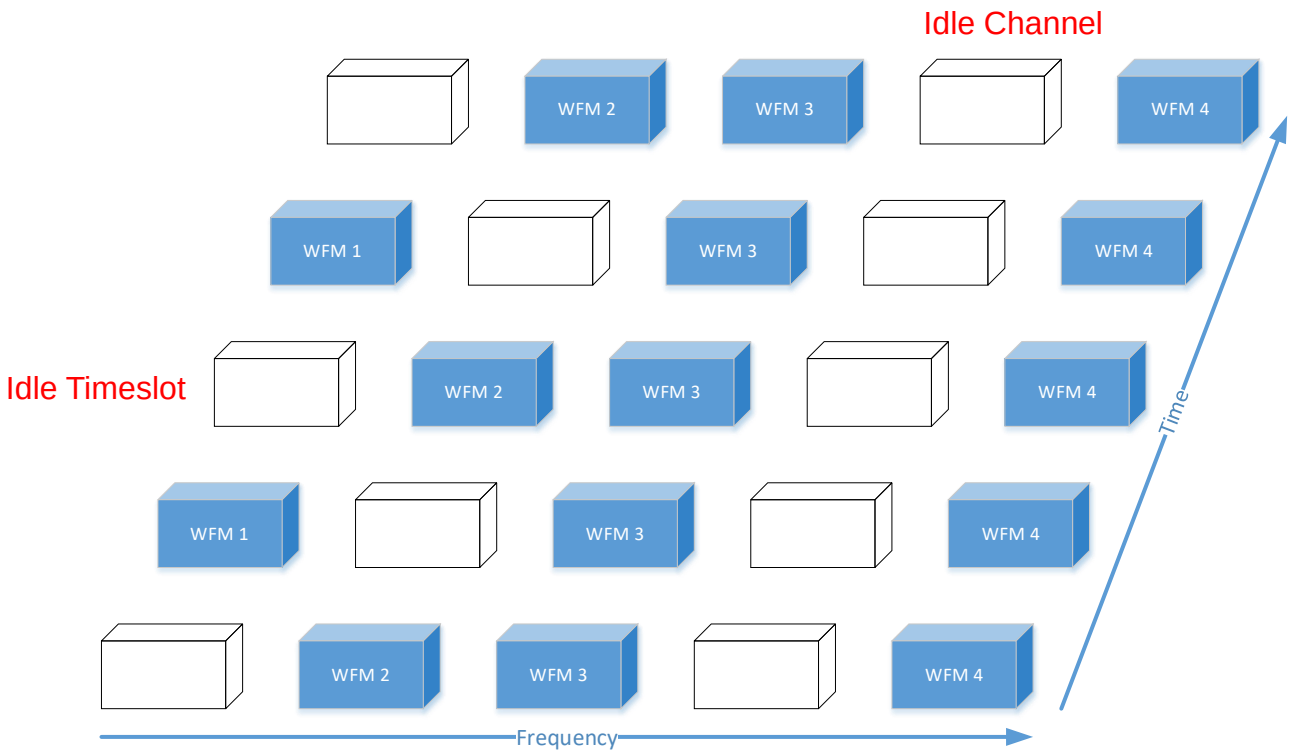
D.Chew, A.Adams, J.Uher, *Wireless Coexistence: Standards and Intelligent Solutions,* Piscataway: IEEE Press, 2021

Secondary Spectrum Access (2)

- SU access types:
  - Underlay – SU accesses resources concurrently, interference negligible or acceptable to PU
  - Overlay – SU accesses resources only when idle (whitespace)
    - Called “Interleaved” in some literature

D.Chew, A.Adams, J.Uher, *Wireless Coexistence: Standards and Intelligent Solutions,* Piscataway: IEEE Press, 2021

Whitespace Example



D.Chew, A.Adams, J.Uher, *Wireless Coexistence: Standards and Intelligent Solutions,* Piscataway: IEEE Press, 2021

Occupancy Determination (1)

- When is the PU active on a given channel?
  - Specific signal testing
    - Use a priori information to detect specific signals
      - Includes database registration
  - Carrier sensing
    - Use energy detection or other methods to sense channel activity
      - PU frame structure not always known

Occupancy Determination (2)

- When is the PU active on a given channel?
  - Predictive models
    - Use historical observations to predict future events
    - Potential benefits
      - Avoids collisions due to erratic PU behavior
      - Avoids collisions due to missed PU detections
      - Increased/prolonged SU traffic
    - Potential challenges
      - Predictions can be wrong
      - Models can be computationally expensive

Background on Markov Models

Markov Models (1)

- Successive observations are often dependent
  - Assume the underlying process is comprised of discrete states rather than independent and identically distributed (IID) random variables
  - Need a model which parameterizes dependencies

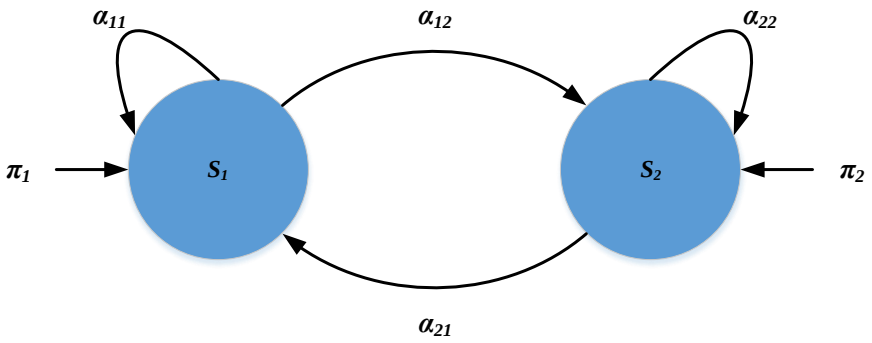
D.Chew, A.Adams, J.Uher, *Wireless Coexistence: Standards and Intelligent Solutions,* Piscataway: IEEE Press, 2021

Markov Models (2)

- Markov model
  - Types of Markov models
    - Markov chain
      - States are observable
    - Hidden Markov model
      - States are a probabilistic function of observations

Markov Models (3)

- Markovian assumption



Two-State Markov Chain

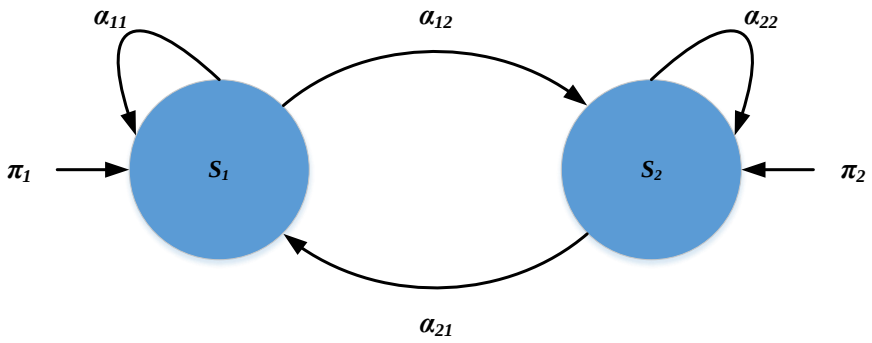
The probability of the next state transition depends only on the current state.

$$P\left(q_{t+1} = S_i | q_t = S_j, q_{t-1} = S_k, q_{t-2} = S_l, \dots\right) = P\left(q_{t+1} = S_i | q_t = S_j\right)$$

D.Chew, A.Adams, J.Uher, *Wireless Coexistence: Standards and Intelligent Solutions,* Piscataway: IEEE Press, 2021

Markov Chain Description (1)

- Model defined by
  - State space $\boldsymbol{S}$
  - Initialization probabilities $\boldsymbol{\pi}$
  - State transition probabilities $\boldsymbol{\alpha}$



Two-State Markov Chain

Transitions are a cumulative product from an initial state

D.Chew, A.Adams, J.Uher, *Wireless Coexistence: Standards and Intelligent Solutions,* Piscataway: IEEE Press, 2021

Markov Chain Description (2)

- The probabilities denoted by α go into a **transition matrix** A

$$
\begin{array}{c|cc}
 & \text{State 1} & \text{State 2} \\
\hline
\text{State 1} & \alpha_{11} & \alpha_{12} \\
\text{State 2} & \alpha_{21} & \alpha_{22}
\end{array}
$$

Rows: Current State

Columns: Next State

α_{11} Probability of staying in State 1 if currently in State 1

α_{12} Probability of transitioning to State 2 if currently in State 1

α_{21} Probability of transitioning to State 1 if currently in State 2

α_{22} Probability of staying in State 2 if currently in State 2

The sum of any row must equal 1

Markov Chain Description (3)

- The sum of any row must equal 1
  - Therefore, we only need two values.

$$\begin{array}{c c} & \begin{array}{cc} \text{State 1} & \text{State 2} \end{array} \\ \begin{array}{c} \text{State 1} \\ \text{State 2} \end{array} & \underbrace{\begin{bmatrix} 1-a & a \\ b & 1-b \end{bmatrix}}_{\text{Columns: Next State}} \end{array} \left.\vphantom{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}\right\} \text{Rows: Current State}$$

$1 - a$ Probability of staying in State 1 if currently in State 1

a Probability of transitioning to State 2 if currently in State 1

b Probability of transitioning to State 1 if currently in State 2

$1 - b$ Probability of staying in State 2 if currently in State 2

Markov Chain Description (4)

- The probabilities of currently being in one of the states is denoted as $\boldsymbol{\pi}$. That is not the number π, just the symbol.

- There are two elements in the vector of probabilities π in the case of a two-state Markov chain.

$$\boldsymbol{\pi} = \begin{bmatrix} \pi_1 & \pi_2 \end{bmatrix}$$

Markov Chain Description (5)

- The probability of being in any state after a number of transitions k is the transition matrix $\boldsymbol{A}$ raised to k.

$$\boldsymbol{A}^k$$

- Each iteration $\boldsymbol{A}^k$ can be calculated as

$$\boldsymbol{A}^k = \frac{1}{(a+b)}\begin{bmatrix} b & a \\ b & a \end{bmatrix} + \frac{(1-a-b)^k}{a+b}\begin{bmatrix} a & -a \\ -b & b \end{bmatrix}$$

- Observe how the transition probabilities change as we look ahead.

Markov Chain Description (6)

- The **Limiting Distribution** is found taking $k \to \infty$

$$\boldsymbol{A}^k = \frac{1}{(a+b)}\begin{bmatrix} b & a \\ b & a \end{bmatrix} + \frac{(1-a-b)^k}{a+b}\begin{bmatrix} a & -a \\ -b & b \end{bmatrix}$$

$$\boldsymbol{A}^* = \frac{1}{(a+b)}\begin{bmatrix} b & a \\ b & a \end{bmatrix} \qquad \boldsymbol{\pi}^* = \begin{bmatrix} \frac{b}{a+b} & \frac{a}{a+b} \end{bmatrix}$$

- The transitions and state probabilities become static as we look further away

- These values can also be calculated by observing the states directly for long periods.
  - *This class of Markov Chain is Irreducible and Ergodic*

Markov Chain Description (7)

- If the Markov Chain is hidden, then we cannot directly observe it.

- We need to rely on events produced by the current state

- Each state represents a different distribution of observable events.
  - This is the **Emission Matrix**

Markov Chain Description (8)

- For the purposes of signal detection, that matrix is

$$B = \begin{bmatrix} 1 - P_{falarm} & P_{falarm} \\ 1 - P_{detect} & P_{detect} \end{bmatrix}$$

Have we observed what we think is a signal?

$$A = \begin{bmatrix} 1 - \alpha_{12} & \alpha_{12} \\ \alpha_{21} & 1 - \alpha_{21} \end{bmatrix}$$

Is there actually a signal present (hidden state)?

α_{12} Probability of transitioning to State 2 if currently in State 1

α_{21} Probability of transitioning to State 1 if currently in State 2

Experiment: Limiting Distribution

Experiment: Limiting Distribution (1)

- Load the script experiment_hmm_limiting_distribution.m
- Review that script
- Execute the script

Experiment: Limiting Distribution (2)

- There are two test cases, one a large average, the other the calculation of the limit
- Are there any differences?
- What could you do with the Limiting Distribution?

Experiment: Predicting Activity

Experiment: Predicting Activity (1)

- Load the script experiment_training_a_hmm_model.m
- Review that script
- Execute the script

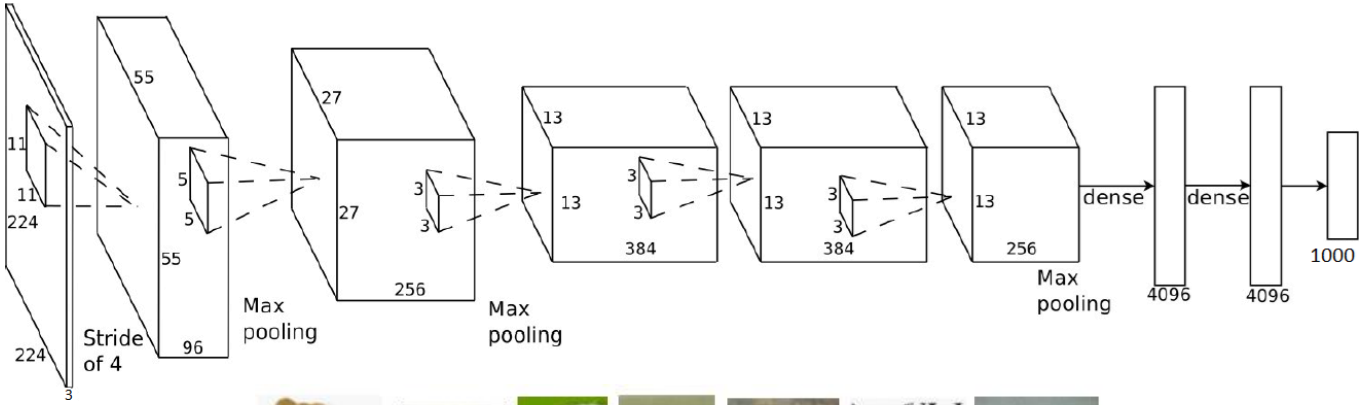
Experiment: Predicting Activity (2)

- There are two test cases, one High SNR, one Low SNR
- What differences did you notice?

Background on Convolutional Neural Nets

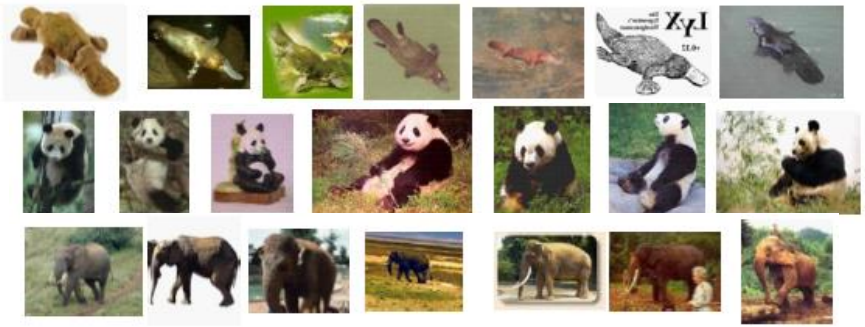
Convolutional Neural Nets (1)

- Convolutional Neural Networks have been widely adopted in Image Recognition problems.



AlexNet

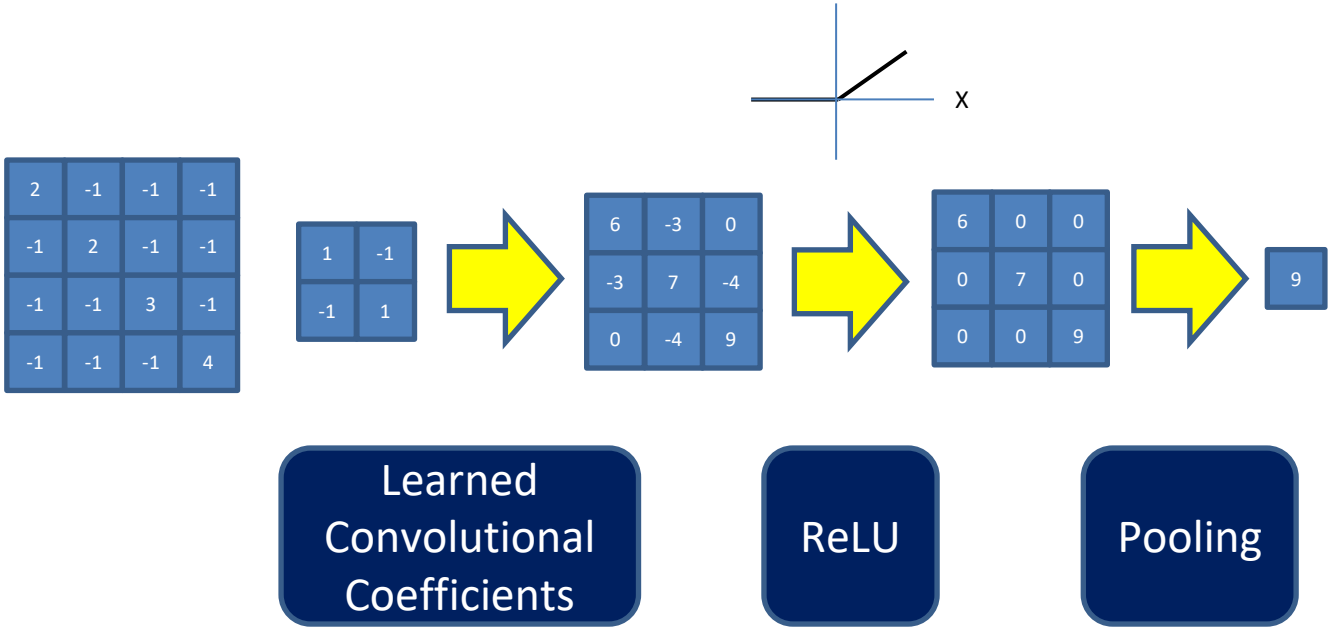
Won ILSVRC 2012



ImageNet Large Scale Visual Recognition Challenge (ILSVRC)

Convolutional Neural Nets (2)

- How do they work?

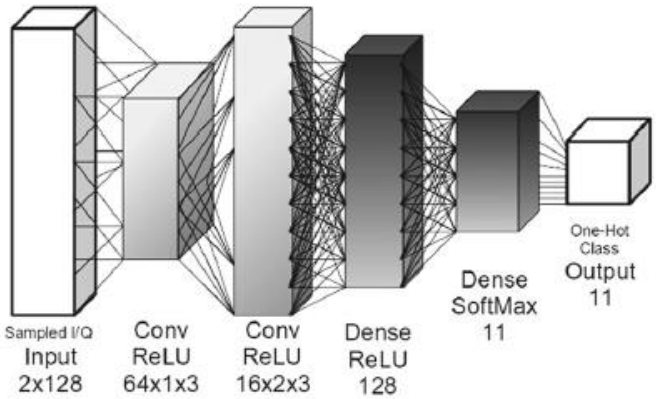


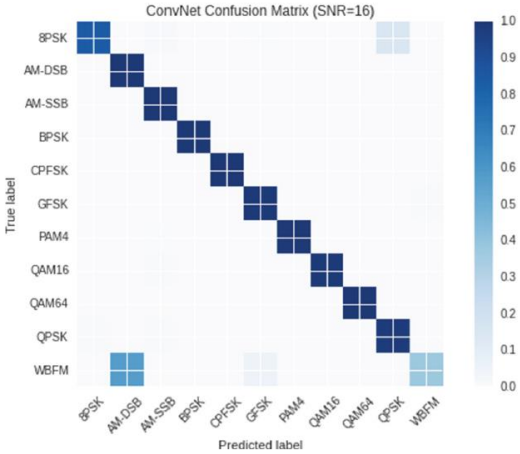
Convolutional Neural Nets for Signals (1)

- Some methods of converting signals into images:
  - Spectrogram
  - Separate real and imaginary components into two columns, with each row being a sample. Then use that whole matrix as an N x 2 image.

Convolutional Neural Nets for Signals (2)

- Such CNNs have been used as modulation classification systems





O'Shea, Timothy J., Johnathan Corgan, and T. Charles Clancy. "Convolutional radio modulation recognition networks." International conference on engineering applications of neural networks. Springer, Cham, 2016

Experiment: Confusion Matrix

Experiment: Confusion Matrix (1)

- Matlab provides a Modulation Classification Network.

- One can train it from scratch using the example from:

  https://www.mathworks.com/help/deeplearning/ug/modulation-classification-with-deep-learning.html

- Alternatively, one can load the provided pre-trained network.
  - That is what we will do!

Experiment: Confusion Matrix (2)

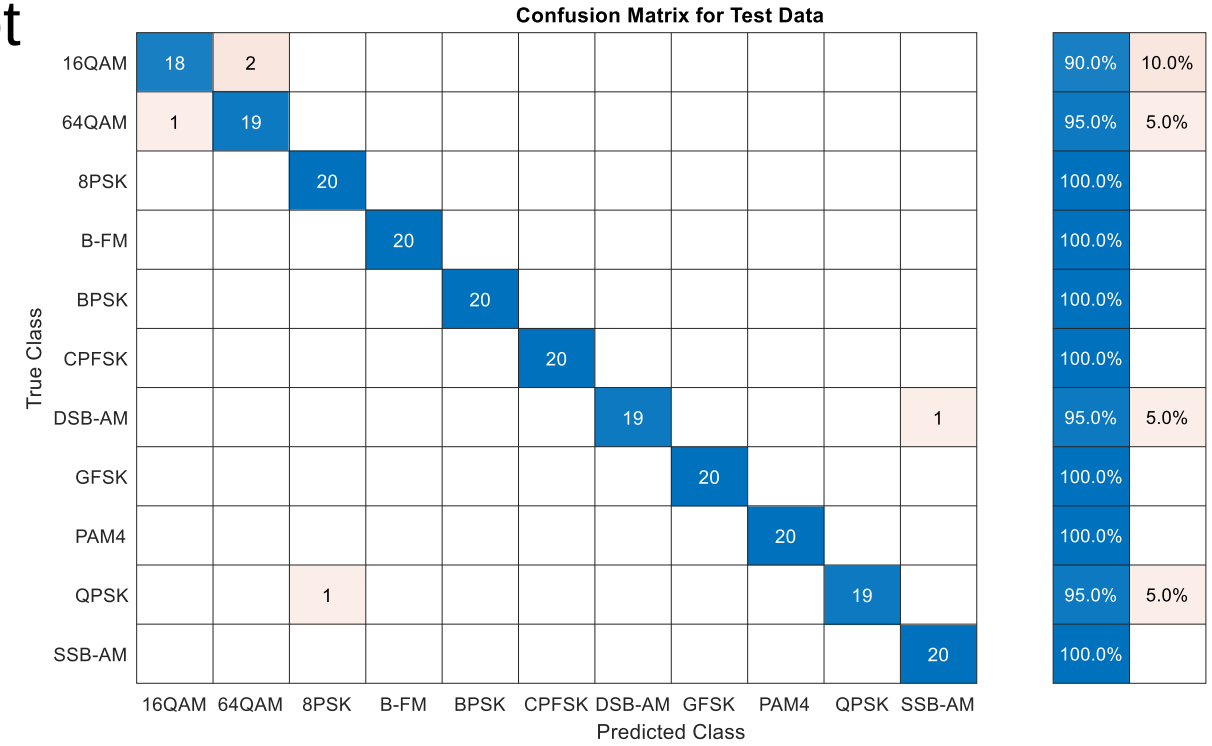
- Load the script modulation_classification_confusion_matrix_experiment.m

- Review that script

Experiment: Confusion Matrix (3)

- Execute the script

modulation_classification_confusion_matrix_experiment.m



Experiment: Noise in Modulation Classification

Experiment: Noise in Modulation Classification (1)

- Load the script modulation_classification_in_noise.m
- Review that script

Experiment: Noise in Modulation Classification (2)

- Execute the script with SNR = 10 dB
  - Observe at or near 100% accuracy
- Execute the script with SNR = 7 dB
  - Is it the same? What happened?
- What does the accuracy look like as a function of SNR?

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